

Code fest - 08

CMAN

Raghuveer Reddy
Vemula.

$N = 1024$ output Neurons
 $f = 50\text{Hz}$ (firing rate)
AER

Task 1 $R = N \times f$

$$N = 1024$$
$$f = 50\text{Hz}$$

$$R = 1024 \times 50$$
$$= 51,200 \text{ spikes/second.}$$

Task 2 Mean required AER BW $| B = R \times 20$.

Each pkt is 10-bit address +
6 bit timestamp + 4 bit parity overhead

$$\text{pkt} = 20 \text{ bits.}$$

$$B = R \times 20$$

$$B = 51,200 \times 20$$
$$= 1,024,000 \text{ bits/sec}$$

Converting to M bits/sec

$$B = 1.024 \text{ Mbit/sec}$$

Task 3 Mean requirement is 1.024 Mb/s

$$\begin{aligned} \text{SPI} & - \leq 50 \text{ Mbit/s} \\ \text{I}^2\text{C} & - \leq 3.4 \text{ Mbit/s} \\ \text{AXI4-like} & - 100 \text{ Mbit/s} \end{aligned}$$

Since the needed bandwidth is less than the given bandwidth of I2C (3.4 Mb/s)

I²C is the least complex interface that suffices.

Task 4 25% of ~~1024~~ 1024 neurons fire within a 1ms window.

$$N_{\text{burst}} = 0.25 \times 1024 = 256 \text{ spikes in } 1 \text{ ms}$$

that means

$$R = \frac{256}{1 \times 10^{-3} \text{ s}} = 256,000 \text{ spikes/s}$$

$$B_{\text{peak}} = R_{\text{peak}} \times 20$$

$$\begin{aligned} &= 256,000 \times 20 \\ &= 5.12 \text{ Mb/s} \end{aligned}$$

Burst to mean R_{mean}

$$\frac{B_{\text{peak}}}{B_{\text{mean}}} = \frac{5.12}{1.024} = 5$$

The earlier chosen interface cannot absorb the 5.12 Mb/s burst instantaneously.

If I²C is still used, a FIFO buffer is needed

The I²C can at most handle

$$3.4 \times 10^6 \text{ bits/sec} \times 10^{-3} \text{ s} / 20 \text{ bits} \\ = 170 \text{ packets.}$$

But we need

$$5.12 \times 10^6 \text{ bits/sec} \times 10^{-3} \text{ s} / 20 \text{ bits} \\ = 256 \text{ packets.}$$

$$= 256 - 170 = 86$$

$\Rightarrow$ A minimum depth of 86 is needed in the buffer as a power of 2 128 depth can be used.

Tasks

$$B_{\text{frame}} = N \times (\text{frames/s}) \times 1 \text{ bit}$$

$$= 1024 \times 1000 \times 1$$

$$= 1.024 \times 10^6 \text{ bits/s}$$

AER BW

$$N \times \text{frames/sec} \times 20 = N \times 1000 \times 1$$

$$f_{\text{crossover}} = \frac{1000}{20} = 50 \text{ Hz}$$

AER/frame ratio @ 50 Hz

$$1.024 \text{ Mb/s} / 1.024 \text{ Mb/s} = 1$$

$$f_{\text{crossover}} = \frac{f_s \times 1}{b} = \frac{f_s}{b}$$

AER is correct choice when the mean firing rate is below the crossover.

Above which the per spike overhead becomes more than the blind sampling of neurons every frame.